

YOLOv8s CV MODEL	0.732 mAP@0.5 (8 classes)	XGBoost PRICING MODEL	19.6% PRICE MAPE	77.6% CONFORMAL COVERAGE	672 TRAINING LISTINGS
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WHAT IT DOES

A user uploads car specs + optional photos. The system runs an **on-device computer-vision damage scan** (no photo ever leaves the browser), prices the car with an **explainable gradient-boosted regression model** calibrated with **conformal prediction**, retrieves real comparable listings via **hybrid retrieval**, and generates a citation-grounded report through an **LLM agent pipeline** that a deterministic Verifier checks number-by-number before it reaches the user.

SYSTEM ARCHITECTURE

Frontend	Next.js 14, TypeScript, Tailwind, Framer Motion, PWA. Runs the CV model client-side via ONNX Runtime Web (WASM execution provider) .
Backend	FastAPI + LangGraph multi-agent orchestration (retrieval agent, aggregation agent, verifier agent), rate-limited + API-key-metered REST API.
Data / Auth	Supabase (Postgres + RLS + Auth), versioned CSV corpus, GitHub Actions cron for scheduled scraping (Apify) and CI evaluation gates.
Hosting	Vercel (frontend), Render (API) — entirely free-tier, zero-downtime deploys with automated rollback on eval regression.

MODELING — DEEP LEARNING vs. CLASSICAL ML (stated explicitly, never blurred)

Deep Learning YOLOv8s (CNN, anchor-free one-stage detector) for 8-class damage detection: dent, scratch, crack, glass_shatter, lamp_broken, tire_flat, punctured, missing_part. - Trained via transfer learning on CarDD + VehiDE (14,437 images), evaluated with mAP@0.5 / mAP@0.5:0.95, IoU-based NMS. - Inference: exported to ONNX, run in-browser via ONNX Runtime Web (WASM) — zero server GPU/RAM cost. - Chatbot: transformer sentence-embeddings (MiniLM) for dense retrieval; teacher-LLM distillation scaffold for a planned on-device LoRA fine-tune.	Classical ML (explicitly NOT deep learning) XGBoost quantile-regression ensemble for price prediction on structured features (make, model, year, mileage, spec, city, etc.) — gradient-boosted trees, not a neural net. Split-conformal prediction for calibrated uncertainty intervals; Mondrian (group-conditional) conformal studied for per-segment coverage. SHAP (native XGBoost <i>pred_contribs</i>) for per-prediction feature attribution shown to the user. Hybrid retrieval RAG: dense embeddings + BM25 + structured similarity, fused and reranked, feeding an LLM (Gemini/Groq) report generator with a deterministic Verifier fact-check gate.
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KEY ENGINEERING FINDINGS (this development cycle)

1	Damage-scoring was area-blind. Both the on-device and server scoring summed detection <i>instances</i> , ignoring bounding-box extent — a crushed front-end and a door-ding both read “1 dent, -2%.” Rebuilt as type-dominant scoring : structural classes (missing_part, punctured, crack) carry real base weight, with co-occurrence escalation for multi-finding collisions.
2	Recall was the real ceiling, not the confidence threshold. Lowering the detector's confidence gate (0.35→0.20) recovered zero additional detections on real crash photos. The actual fix was tiled inference — full frame + 3 overlapping crops — which took one wreck from 1 detection to 11.
3	Weighted Box Fusion (WBF) replaced per-class NMS for tile-merging: instead of discarding overlapping boxes, WBF fuses them into a confidence-weighted average with an agreement bonus when independent crops concur — tighter localization, more trustworthy confidence.
4	Pixel-based severity head. Severity is now graded from the detected crop's own pixels — gradient/texture energy, dark-pixel fraction, extent — on a 48×48 grayscale patch, not inferred from box size. Verified byte-identical between the TypeScript (browser) and Python (server) implementations.
5	Pixel evidence can rescue low-confidence detections that the raw model confidence would otherwise suppress — a severe-looking crop at 0.35 confidence is trusted more than an identical-confidence but visually mild one.
6	glass_shatter is the model's dominant false-positive class — sky/scene reflections on an intact windshield are indistinguishable from real shatter on texture, edge-density, <i>and</i> saturation (all three tested, all three overlap). Mitigated, not eliminated, by down-weighting its severity and raising its confidence gate — true fix requires retraining with reflection hard-negatives.

HONEST EVALUATION (the project's stated design principle)

Metric	Result	Context
CV detection (mAP@0.5)	0.732	Beats published CarDD benchmark (≈0.58); dent/crack recall is the known weak class (0.37–0.53)
Price prediction (MAPE)	19.6%	SOTA on large clean corpora is ≈7.6% — gap is data volume (672 rows), a stated, tracked limitation
Conformal interval coverage	77.6%	Target 80%; corpus-bound, re-evaluated as the training corpus grows
RAG retrieval (same-make P@5)	0.780	Proven at the mathematical ceiling the corpus permits — formally retired as a tuning target

Design principle: never overstate accuracy or confidence. Every valuation ships with its interval, its coverage guarantee, its SHAP attribution, and its comparable citations — a wrong-but-honest number the user can verify beats a confident-but-fabricated one.